Lennart Bedarf

PhD Candidate in Biomedical Engineering

University of Basel · qMRI · MR spectroscopy · postmortem imaging



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German · born 01.09.1999
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Languages: German, English

EDUCATION

06/2024 – present
PhD in Biomedical Engineering
University of Basel · Basel, Switzerland
Thesis: Multimodal Brain Tissue Analysis: From Forensic Temperature-Based Postmortem Interval Estimation to Advanced qMRI Anisotropy and Histological Validation Techniques

- Day-to-day supervision of Master's students.
- Lecture on *Forensic Imaging (MRI)* (annually since 2025, ∼ 30 students, Master's level).

04/2022 – 01/2024
M.Eng. in Medical Engineering **Grade 1.0, with honours**
South Westphalia University of Applied Sciences · Hagen, Germany
Focus: advanced imaging techniques in MRI.
Master's thesis (1.0): "Studies on edited magnetic resonance spectroscopy: Phantom- and *in vivo*-measurements for neuroscientific problems"

09/2018 – 04/2022
B.Eng. in Medical Engineering **Grade 1.5**
South Westphalia University of Applied Sciences · Hagen, Germany
Bachelor's thesis (1.0): "Python-based analysis and evaluation of spectroscopic *in vivo* MR data"

2010 – 2018
Abitur (general university entrance qualification)
Zeppelin-Gymnasium · Lüdenscheid, Germany

WORK EXPERIENCE

12/2022 – 06/2024
Working Student
TEBIT GmbH & Co. KG · Meinerzhagen, Germany
Part-time role in high-precision medical engineering alongside the M.Eng. programme.

2020 – 2022
COVID-19 Tester (part-time)
Johanniter Seniorenheime GmbH · Lüdenscheid, Germany
Conducted rapid antigen tests for visitors of a nursing home.

PUBLICATIONS

Journal Articles

2026
L. Bedarf, K. Gerlach, H. Wittig, E. Scheurer, C. Berger, C. Lenz: "Evaluating the forehead temperature for estimating the postmortem interval". *International Journal of Legal Medicine*, 2026. doi:10.1007/s00414-026-03736-x

2025
J. Bottoni, H. Wittig, T. Rost, A. Schocker, P. Wild, U. Nachbur, D. Neuhaus, **L. Bedarf**, K. Gerlach, E. Scheurer, C. Lenz: "Detection of gunshot residues using infrared photography: influence of ammunition type, surface color and blood contamination". *International Journal of Legal Medicine*, 2025. doi:10.1007/s00414-025-03609-9

2024
H. Nawrath, A. Holl, **L. Bedarf**, D. Mai, H. Alfke, J. Gröbner: "Optimized Schlieren imaging for real-time visualization of high-intensity focused ultrasound waves". *Biomedical Engineering / Biomedizinische Technik*, 2024. doi:10.1515/bmt-2024-0002

Conference Abstracts

2026
L. Bedarf, D. Neuhaus, E. Scheurer, C. Lenz: "Incomplete reversibility of fixation-induced changes in postmortem human brain white matter MRI following rehydration". *ESMRMB Annual Meeting*, Girona, 2026.

2026
L. Bedarf, D. Neuhaus, E. Scheurer, C. Lenz: "Fixation and rehydration influence postmortem MRI parameters". *Postmortem Research in Dementia Workshop*, New York, 2026.

2026 **L. Bedarf**, K. Gerlach, H. Wittig, E. Scheurer, C. Lenz: “Time of death estimation using forehead temperature measurements”. *IAFS Triennial Meeting*, Sofia, 2026.

2025 **L. Bedarf**, D. Neuhaus, M. J. Wendebourg, R. Schlaeger, C. Birkl, E. Scheurer, C. Lenz: “Quantifying the impact of formalin fixation on R_2^* orientation dependency in human brain white matter” (poster PG253). *ESMRMB Annual Meeting*, Marseille, 2025.

2023 **L. Bedarf**, A. Holl, H. Nawrath: “Determination of T_1 and T_2 relaxation times of phosphorus-31 in a low-field MRI by developing a phosphorus-31 resonant low-field MRI coil”. *Proc. International Society for Magnetic Resonance in Medicine (ISMRM)*, 2023.

AWARDS & FUNDING

2026 **Best Poster Prize (2nd place)**, ESMRMB 42nd Annual Scientific Meeting, Girona, Spain – EUR 100

2026 **Best Poster Prize**, ESMRMB Microstructural Imaging Workshop, Girona, Spain – EUR 100

2026 **Trainee Travel Award**, Postmortem Research in Dementia Workshop, New York, USA – USD 200

2026 **Travel Grant** (*Reisefonds*), University of Basel, for the IAFS Triennial Meeting in Sofia – CHF 1,101

09/2021 – 08/2023 **Deutschlandstipendium**, merit-based scholarship for outstanding academic performance, Federal Ministry of Education and Research (BMBF)

ACADEMIC SERVICE & TRAINING

2025 **Chairperson**, ESMRMB Annual Meeting, Marseille

2025 **University of Basel training courses:** Presentation Training (02/2025); Good Scientific Practice and Its Pitfalls in Everyday Research (06/2025); Self-Branding and Self-Promotion (10/2025); Stress Management in Performance Situations (10/2025)

SKILLS & INTERESTS

Languages German (native) · English (fluent) · Latin (basic, *Kleines Latinum*)

Programming Python, MATLAB, C

Engineering CAD, SolidWorks

Tools Linux, Git, LaTeX, Microsoft Office

Interests Handball (since 2004) · Cooking (catered for 30 people at a week-long youth camp) · Volunteering, Evangelische Versöhnungskirchengemeinde Lüdenscheid · Successful participation in the international “Formula 1 in Schools” competition

Other Driving licence, class B